

Parikshit Kumar

mr.kumar.parikshit@gmail.com • [LinkedIn](#) • [GitHub](#) • +1 (951)-544-4046 • California, USA

SKILLS

Technical Skills: Deep Learning, NLP, LLMs, Computer Vision, Generative Modeling, Diffusion Models, Machine Learning, Reinforcement Learning, Data Analysis, Data Visualization, Statistics, Time Series Analysis, Transfer Learning, Model Optimization, Python, C++, C, Keras, TensorFlow, PyTorch, PyTorch Lightning, FastAI, OpenCV, Numpy, Matplotlib, Pandas, Hugging Face, SQL, Flask, BeautifulSoup, Graph Neural Networks, XGBoost

Tools & Platforms: Tableau, MLOps, GCP, AWS, Kubernetes, Docker, Git CLI, Jupyter Notebooks, Anaconda, Linux Terminal, Robot Operating System (ROS), CUDA, MPS (Apple), Apache Spark, Tensorboard, Firebase, MS Azure

EXPERIENCE

Junior Specialist - *University of California, Riverside* Feb 2025 - Present

- Developed deep learning models for mosquito orientation classification with **96% accuracy** across 4 orientations, enhancing morphological pattern detection.
- Applied advanced image processing (thresholding, edge detection, morphological transformations) for robust biological feature extraction.
- Collaborated with interdisciplinary teams to integrate model deployment pipelines, optimizing end-to-end performance and scalability.

Student Assistant - *University of California, Riverside* Jul 2024 - Dec 2024

- Conducted sensitivity analysis with a **50% sample size increase**, reducing variance by **20%** and boosting detection accuracy by **30%**.
- Built integrated pipelines merging biological datasets with machine learning models, cutting feature extraction time by **40%**.
- Collaborated with entomologists to refine morphological pattern identification for enhanced interpretability.

Research Intern - *University of Guelph, Ontario* Jan 2023 - Jun 2023

- Applied image segmentation and LLM-based text analysis to assess soil properties, achieving **18% accuracy improvement**.
- Built automated analysis applications reducing data processing time by **40%** and improving health indicator detection by **25%**.
- Co-authored a research paper on data-driven soil characterization, contributing to a **15% performance increase** and enhanced decision-making.

Computer Vision Intern - *UT Dallas & IIT Bombay* Mar 2022 - Jul 2022

- Implemented Faster R-CNN and YOLO for autonomous vehicle applications, increasing object detection accuracy by **15%**.
- Enhanced off-road segmentation accuracy by **20%** using 3D simulations and generative models.
- Streamlined large-scale data preprocessing and camera calibration, reducing training time by **25%** and boosting feature detection precision by **18%**.

Machine Learning Intern - *Indian Institute of Technology, Bombay* Jan 2021 - Sep 2022

- Developed ML/NLP knowledge curation engines that improved search relevancy and reduced query response times by **35%**.
- Automated anomaly detection in large-scale datasets, achieving a **20% reduction** in manual oversight.
- Optimized and streamlined model pipelines using Python and SQL to enhance computational efficiency.

EDUCATION

Master of Science in Computer Science, GPA: 3.6 - *UC Riverside* Sep 2023 - Dec 2024

Bachelor of Technology in Computer Science, GPA: 4.0 - *SRM IST* Jun 2019 - May 2023

PROJECTS

Generative Networks and VAEs to Assist Artists - *SRM IST, Paper and Project*

- Engineered variational autoencoders achieving **90.6% accuracy** on the CUHK Face Sketch Database, enhancing artist workflow efficiency by **30%** and presenting at ICCET 2023.

Energy Price Forecasting Using LSTMs and ARIMA - *SATU, Paper and Project*

- Collaborated on energy price forecasting using LSTMs and ARIMA, increasing prediction accuracy by **18%** for Australian datasets and presenting findings at AIJR Conference 2022.